

Where Should Ghana's Nets Go?

Data-driven allocation of insecticide-treated nets ·
ICS553 Prosit 1

Group 4

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A limited delivery of nets has arrived. Where will it do the most good, most fairly and how sure can we be?

Proposal to the Ministry of Health allocation committee

The decision in front of the committee

Nets are limited, so every choice is also a choice about who waits. We had to defend four things:

1

The estimate

Malaria cases per 100,000 people, not raw counts

2

The uncertainty

Honest ranges that respect how the survey was collected

3

The honesty checks

No sampling bias and no data leakage

4

The decision

Which areas come first, and why

Our data, and what's missing

50 of 260

Ghana's districts have district-level case data

3 of 16

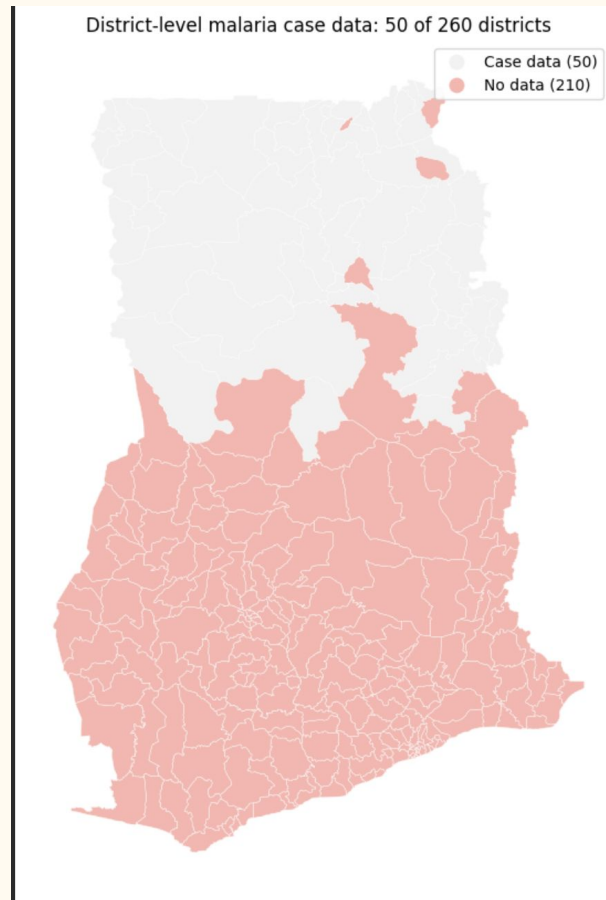
regions covered: Northern, Upper East, Upper West

0

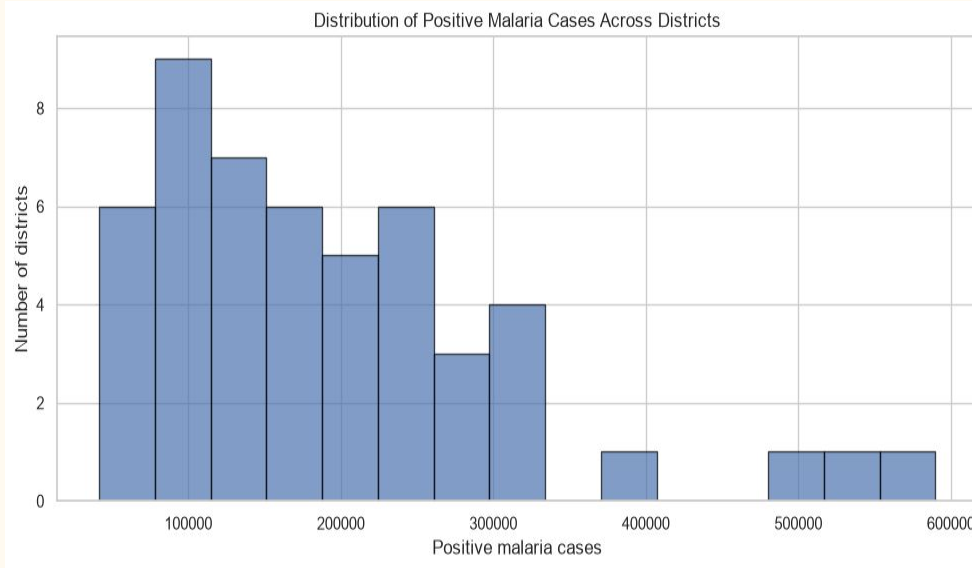
missing cells in either file, and every district reported all 48 months

The brief vs the data: the district file has no rainfall, temperature or elevation, and the survey has no district or prevalence column.

Sources: *ghana_district_cases.csv*; *gha_admin2.geojson*; *data_dictionary.md*.



The counts are extremely spread out



Right-skewed

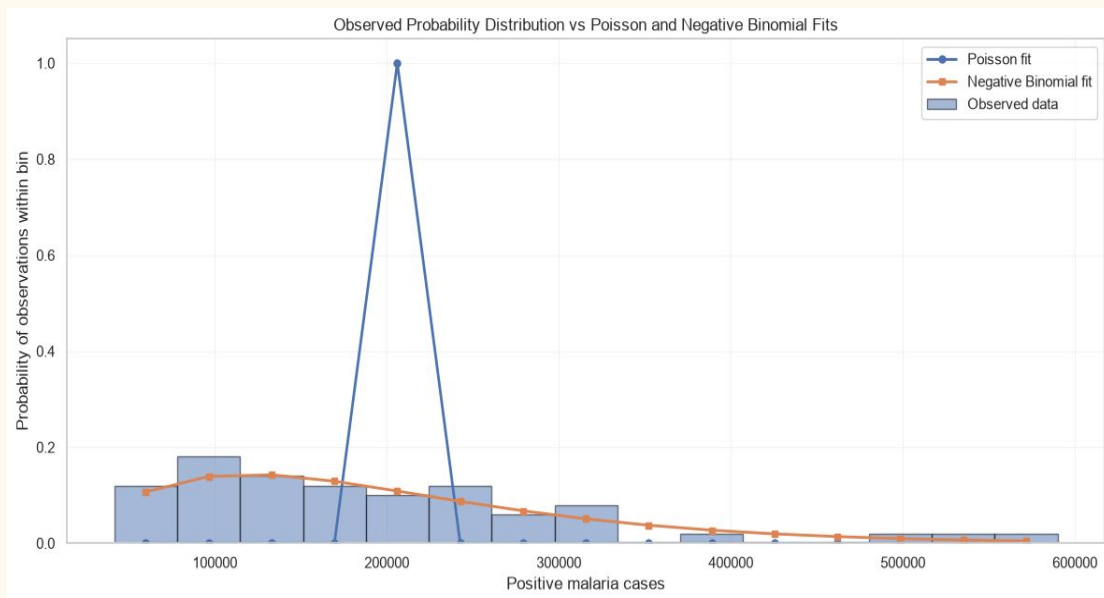
most districts are low; a few reach
500,000–600,000 cases

$\approx 77,000$

variance $\div$ mean. A Poisson model needs this to
be 1.

Mean **195,640** cases, variance **15.1 billion**.
Districts differ by about 123,000 cases; a
Poisson would allow about 442.

Poisson fails; the Negative Binomial fits



$$\alpha = 0.3416$$

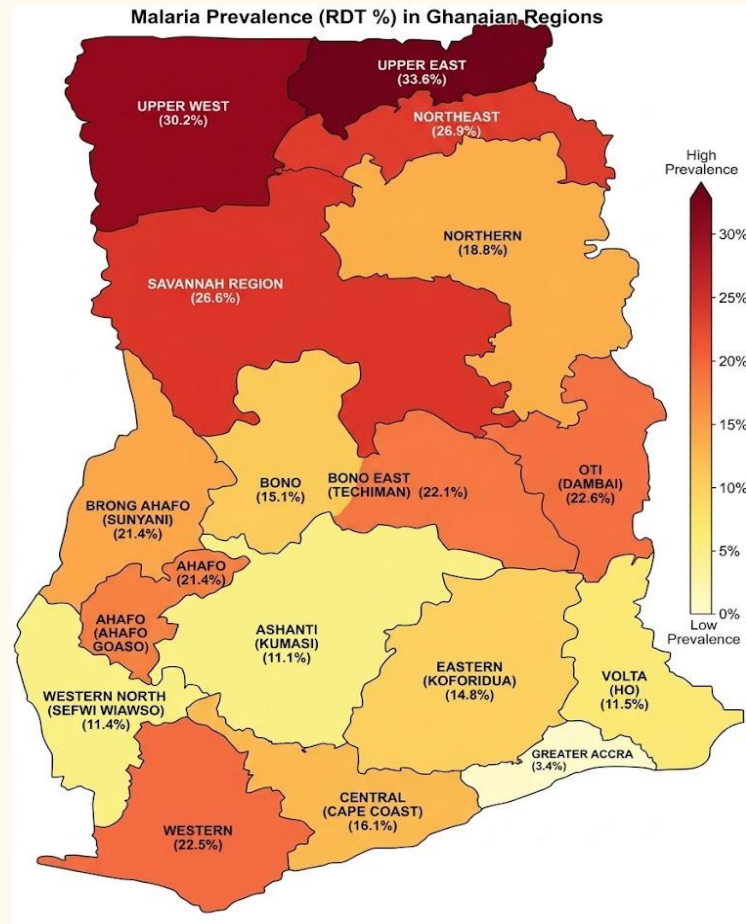
extra-spread parameter (statsmodels, maximum likelihood); range 0.215–0.469 excludes zero

Variance: data 15.1 bn · Negative Binomial 13.1 bn · Poisson 195,640

The Poisson piles all its probability into one spike. The Negative Binomial follows the real shape of the data.

Curves: method-of-moments fit ($\alpha \approx 0.39$), which agrees closely with the maximum-likelihood $\alpha = 0.3416$.

Look Up There



The national picture

Chance each region's child malaria prevalence is above 25%:

Tier 1

Upper East 99% · Upper West 89% · North East 73% · Savannah 65%

Tier 2

Oti 31% · Western 28% · Bono East 23% · Ahafo 11%

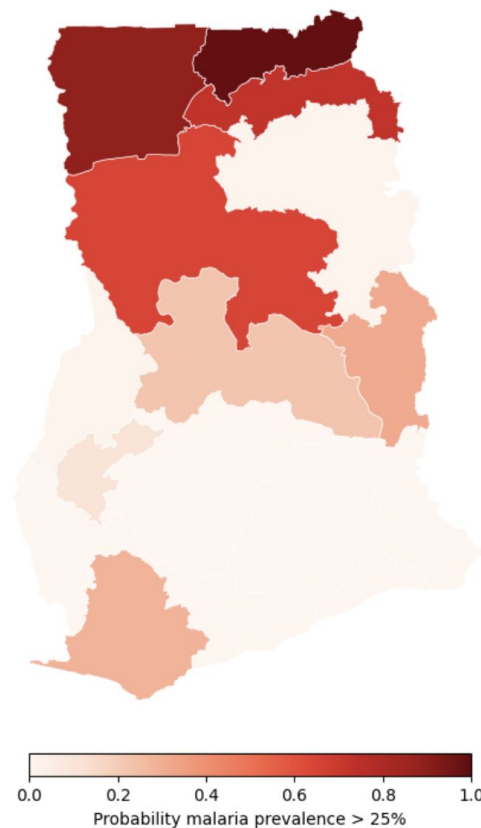
Tier 3

Northern, Bono, Central, Eastern, Volta, Western North, Ashanti, Greater Accra: under 2%

The north is almost certainly above 25%. The middle belt can't be ruled out. The south is confidently lower.

Source: 2022 DHS subnational indicators (RDT, children 6–59 months). Probability approximated from published 95% CIs.

Chance each region's child malaria prevalence exceeds 25% (2022)

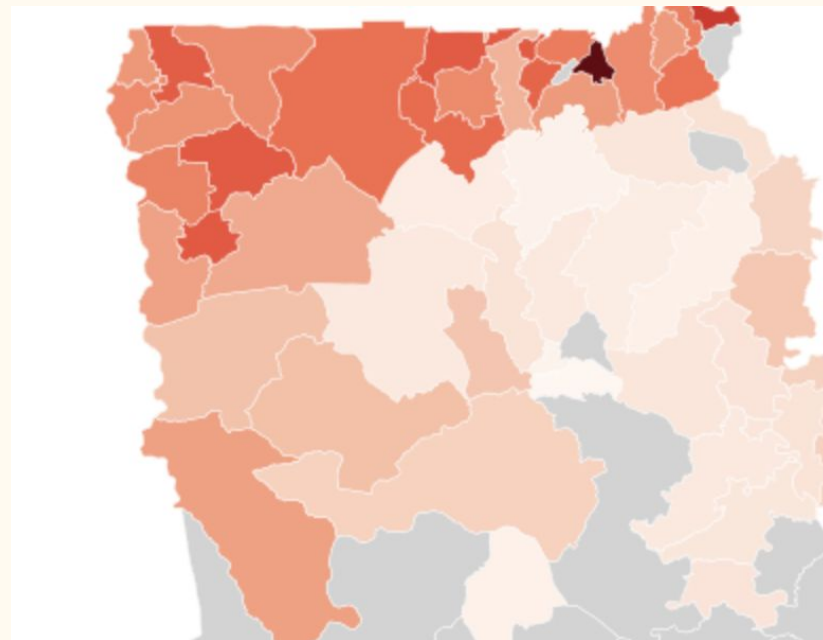


Where malaria is highest

Upper East carries the heaviest burden, followed by Upper West.

District	Region	Cases per 100k
Nabdam	Upper East	713,356
Pusiga	Upper East	507,004
Wa	Upper West	427,551
Daffiama-Bussie-Issa	Upper West	423,544
Kasena-Nankana West	Upper East	420,790

Rates above 100,000 aren't errors: they add up four years (2014–17) and count repeat infections. Read them as relative burden between districts.



Two sources disagree

2022 survey

Children 6–59 months, tested at home

North East: 26.9%, 73% chance above 25%

Savannah: 26.6%, 65% chance above 25%

Both are Tier 1

2014–17 clinic records

Confirmed cases at clinics, all ages

Highest district, Bole: 270,173 per 100k

Every top Upper West district is above 400,000

Both regions look modest

What we conclude: the reason for the gap isn't measured in our data. One possibility is limited access to clinics, which would mean fewer cases get recorded. We don't deprioritise these regions on clinic records alone.

Also: 11 of the 26 “Northern” districts in the 2014–17 data now belong to Savannah or North East, regions created in 2019.

How sure are we?

Net ownership among rural households in Northern Region: 582 households, but only 20 villages.

Resampling households (ignores villages)

70.4% – 77.5% · 7.0 points wide

Resampling whole villages (honest)

66.1% – 81.9% · 15.8 points wide

| = estimate 74.1%

60% 65% 70% 75% 80% 85%

2×

wider once we respect how the survey was collected

76.0%

weighted estimate, using survey sample weights (unweighted 74.1%)

Households in one village are alike. Ignoring that makes us look twice as certain as we are.

Keeping ourselves honest

0.785

R^2 using leaky columns. Looks like a good model.

-0.22

R^2 without them. Worse than guessing the average.

Same linear model, 40 training and 10 test districts, split stratified by region. Only 10 test districts, so treat the gap as illustrative.

Leak	Why it's a problem
Time: net coverage	Measured in 2022, after the 2014–17 cases
Target: cases per 100k	Calculated directly from positive cases
Near-target: tested, suspected	Recorded at the same visits; known only afterwards
Preprocessing on all data	Lets the hidden test set shape the training data
Survey clusters	Neighbours in train and test are near-copies

No missing values needed filling. Only `region_name` was one-hot encoded; target encoding was avoided.

Our recommendation

Priority	Where	Why
1	Upper West: Wa, Daffiama-Bussie-Issa, Lambussie-Karni first	High malaria in both sources, and lower net ownership (69.8%)
2	Savannah and North East: Bole, West Gonja, Sawla-Tuna-Kalba first	Tier 1 in the 2022 survey, but low in clinic records, so weaker evidence
3	Tier 2 regions: Oti, Western, Bono East, Ahafo	High need can't be ruled out; a smaller share, plus a request for district data
Not nets first	Upper East	Highest malaria, but highest net ownership (79.6%): education on net use, and newer case data
Lowest	Tier 3 regions	Lower need, with more confidence; Greater Accra most confidently lowest

Within a group, districts with close rates aren't ranked against each other. Wa's rate may include patients from other districts.

Limitations and next steps

What limits our confidence

Only 50 of 260 districts have case data

Case data (2014–17) is older than net data (2022)

Net ownership is only by region; 11 districts carry the wrong region's figure

Only 10 test districts in the leakage check

The probability map assumes symmetric ranges

What would help next

District-level case data for all 16 regions

Case data after 2022, to see whether Upper East's nets are working

Net use, not just ownership (children under a net last night)

Re-running this ranking before the next delivery

AI Declaration

AI use: AI supported concept learning, starter code and slide preparation; all analysis was run, checked and interpreted by the team. Licensed survey data was not shared with any AI tool.



Thank you. Questions?